

Glass Classification using KNN, Decision Trees, and Naïve Bayes

Introduction

Glass classification is motivated by criminal investigations to identify glass found at crime scenes, so that it can be used as forensic evidence. The goal is to classify glass samples into one of six types based on its chemical composition, provided by the glass identification dataset. This study implements three classifiers: K-Nearest Neighbour (KNN), Decision Trees (DT), and Naïve Bayes (NB), evaluated on accuracy, speed, and confusion matrices.

Data and Preparation

The dataset contains 214 samples across 6 glass types: 1, 2, 3, 5, 6, 7 with 9 features: Refractive index (RI), Na, Mg, Al, Si, K, Ca, Ba, Fe. The ID column was removed, and data was split 80/20 into training and test sets using a random permutation seeded on my student ID to ensure reproducibility. As KNN is sensitive to scale differences between features, standardisation using StandardScaler was used fitted on only the training set.

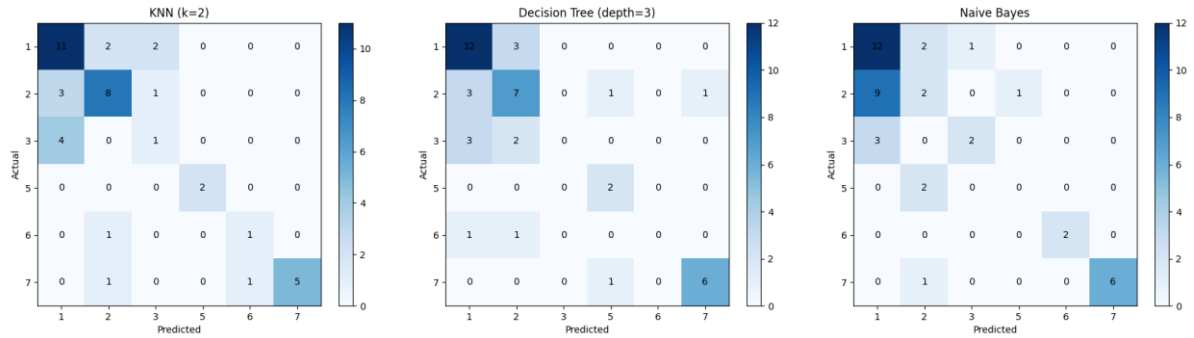
Methodology

KNN classifies samples by the majority class of the nearest neighbours based on Euclidean distance, with the K value tuned in a range from 1 to 4 to find the best K value. DT builds a tree recursively to split data by feature thresholds that minimise Gini Impurity. Maximum depth was tuned in a range from 1 to 4 to avoid overfitting. NB calculates the likelihood of each feature, assuming features are independent, and posterior probability is calculated to identify most likely class.

Results

Model	Config	Accuracy	Time Taken
KNN	k=1	62.79%	0.035s
KNN	k=2	67.44%	0.039s
KNN	k=3	67.44%	0.041s
KNN	k=4	65.12%	0.036s
DT	depth=1	39.53%	0.031s
DT	depth=2	60.47%	0.057s
DT	depth=3	62.79%	0.090s
DT	depth=4	55.81%	0.119s
Naive Bayes	-	55.81%	0.003s

KNN performed best overall at k=2 and k=3 with 67.44% accuracy in 0.04s. It handled majority classes 1 and 2 well, but struggled with minority classes 3, 5, and 6 as they had fewer training samples. DT had best accuracy (62.79%) at depth=3, but accuracy dropped after depth=4 suggesting overfitting. Time also grew from 0.031s at depth=1 to 0.119s at depth=4. Class 3 was frequently falsely identified as classes 1 and 2. NB achieved 55.81% accuracy but was by far the fastest model at 0.003s. It consistently misclassified class 2 instances as class 1, likely due to features being correlated when NB assumes features are independent.



Conclusion

KNN at $k=2$ is the recommended model, as it achieved the highest accuracy (67.44%) whilst being consistently fast (0.04s). It suits the dataset well as it makes no assumptions about the data. Its limitations are that predictions are made against every training sample, which becomes computationally expensive as the dataset grows. Further research could use feature selection to reduce noise, as well as collecting more samples of minority classes 3, 5, and 6 as KNN struggled most.